

 What is a Number Sequence?

A **sequence** is an ordered list of numbers that follows a secret rule! Each number in the line is called a **Term**.

- **1st Term:** The very first number.
- **Term-to-Term Rule:** The step used to hop to the next number.

 Stepwell (Baoli) Hop!

Imagine hopping down steps in an ancient stepwell skipping every 2 steps!

 Growing & Shrinking Patterns Growing Pattern (Adding/Multiplying)

Numbers get **bigger**!

Example: Aarav saves in his Gullak (Piggy Bank):

Day 1: ₹10 | Day 2: ₹15 | Day 3: ₹20

Rule: **Add ₹5 each day (+5)**

 Shrinking Pattern (Subtracting)

Numbers get **smaller**!

Example: Tender Coconut vendor selling stock:

Start: 50 | 1 Hour: 44 | 2 Hours: 38

Rule: **Subtract 6 coconuts (-6)**

 The Magic Pattern Machine (Position Rules)

Want to find the **10th term** without calculating all terms in between? Use the **Position $\times$ Rule** shortcut!

Example: Rangoli Flower Petals 

- Design 1 (1st Position): $1 \times 4 = 4$ petals
- Design 2 (2nd Position): $2 \times 4 = 8$ petals
- Design 3 (3rd Position): $3 \times 4 = 12$ petals
- **Shortcut Secret!** How many petals in Design 10? $\rightarrow 10 \times 4 = 40$ petals!

 Spotting the Impostor (Finding Errors)

Check if every jump follows the exact same rule! Look at these Auto-rickshaw fares:



Busted! ₹36 breaks the "+5" rule! It should be ₹35!



Student Practice Worksheet



Grade 4 Math • Session 1: Number Patterns & Rule Discovery

Name: _____

Date: _____

Score: ____ / 14

Question 1: What is the term-to-term rule for this sequence? **4, 8, 12, 16, 20...**

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Question 2: Is the sequence **85, 75, 65, 55...** a *Growing* pattern or a *Shrinking* pattern?

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Question 3: Fill in the missing term in the sequence: **7, 14, 21, ___, 35**

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Question 4: Kabir has ₹10 on Day 1, ₹18 on Day 2, and ₹26 on Day 3. How much money will he have on Day 4?

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Question 5: A coconut vendor starts with 60 coconuts and sells 7 coconuts every hour. What are the next 3 terms?
60, 53, 46, ___, ___, ___

.....

Question 6: Find the term-to-term rule for this multiplying pattern: **3, 6, 12, 24, 48...**

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Question 7: Spot the Impostor! One number in this sequence is wrong. Find and correct it:
15, 20, 25, 31, 35, 40

.....

Question 8: Ananya makes Rangoli designs. Design 1 has 5 petals, Design 2 has 10 petals, and Design 3 has 15 petals. Using the Position $\times$ Rule shortcut, how many petals will be in **Design 8**?

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Question 9: A Delhi Metro train arrives at the station every 8 minutes starting at 8:00 AM. Write the arrival times for the next 3 trains:
8:00 AM, 8:08 AM, _____, _____, _____

.....

Question 10: What is the 1st term and the 5th term in a sequence that starts at **9** and follows the rule "Add 6"?

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Question 11: True or False: In a shrinking pattern, you always perform addition or multiplication.

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Question 12: Diya lights Diwali lamps (Diyas). Table 1 has 3 diyas, Table 2 has 6 diyas, Table 3 has 9 diyas. What is the rule linking the Table Number (Position) to the number of Diyas?

Question 13: Calculate the **12th term** in the 6-times table skip counting sequence (6, 12, 18...) without writing all the terms!

Question 14: Fill in the missing numbers in this double-step sequence: **100, 92, __, 76, __, 60**
